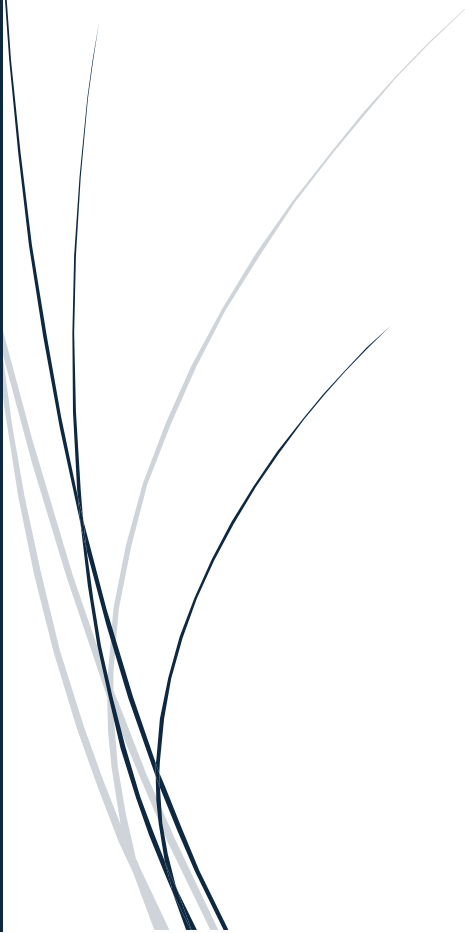




ACTL3182

Report

Projects 1 and 2



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Project 1

1.1

For our investigation, we selected Stockland Group (SGP), Computershare Limited (CPU), and Medibank Private Limited (MPL) as they operate in three distinct sectors being real estate, financial services, and healthcare, respectively. We made this selection with the goal of attaining diversification benefits and reducing exposure to sector-specific risks.

All three companies were chosen from the Australian market (ASX) to ensure consistency in data format, trading calendars, and market holidays. This uniformity would enhance the accuracy and comparability of the analysis. However, it is important to note that although the portfolio achieves sectoral diversification, the companies' shared geographical exposure to Australia fails to address the geographical, country-specific risk.

In our modelling, the analysis period begins effectively on 15 July 2022, even though the nominal start date is 4 July 2022. Since 4 July 2022 fell on a Monday, the last trading day of that week was Friday, 8 July 2022, and since we are working with weekly returns, the first complete weekly return is measured between 8 July 2022 and 15 July 2022, hence, we set the return series to begin on 15 July 2022. While such date adjustments may seem minor, accounting for these details improves the reliability and robustness of the results.

We used Excel for our calculations. To begin, we imported the weekly adjusted stock prices for the three companies, then found the arithmetic return:

$$r_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}$$

We found the weekly risk-free return using Bank Accepted Bills (BABs) because they are a common proxy used in Australia, as they are short-term, low-risk money market instruments issued by highly rated banks. Therefore, they are considered to carry minimal default risk and are highly liquid. Additionally, unlike government bonds, which have longer maturities and can be affected by term premiums and liquidity effects, BABs generally mature within 30 to 180 days, making them well-suited to short-term investment strategies. We found the weekly risk-free return:

$$r_{weekly} = (1 + r_{annual})^{\frac{1}{52}} - 1$$

We found the mean, variance, correlation and covariance of the assets in Excel so that we could use that information to ultimately optimise the weight allocation to the risky assets in the portfolio. We used sample mean, sample variance, and sample covariance rather than population measures because the data set only covers a limited historical period.

$$E(r_i) = \frac{\sum r}{n} = \mu_i$$
$$Var(r_i) = \frac{\sum (x_i - \bar{x})^2}{n - 1} = \sigma_i$$
$$cov(r_i, r_j) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1} = \sigma_{ij}$$
$$corr(r_i, r_j) = \frac{cov(r_i, r_j)}{\sigma_i \sigma_j} = \rho_{ij}$$

The risk-free asset is excluded from the covariance and correlation matrix because it has zero variance. With no variance, it also has no covariance with any risky asset.

Table 1: Mean and standard deviation of all assets

	SGP	CPU	MPL	Risk Free
Mean	0.3984%	0.4098%	0.3760%	0.0623%
Standard Deviation	0.0793%	0.1306%	0.0759%	0%

Table 2: Correlation Matrix of risky assets

	SGP	CPU	MPL
SGP	1	0.0402	-0.0259
CPU	0.0402	1	0.1418
MPL	-0.0259	0.1418	1

Table 3: Covariance Matrix of risky assets

	SGP	CPU	MPL
SGP	0.0793%	0.0041%	-0.0020%
CPU	0.0041%	0.1306%	0.0141%
MPL	-0.0020%	0.0141%	0.0759%

- The three risky assets exhibit relatively small differences in mean weekly returns, with CPU having the highest average weekly return (0.4098%) and MPL the lowest (0.3760%).
- However, the standard deviations do have meaningful variation, CPU is the riskiest asset and MPL and SGP are considerably less volatile, as seen in Table 1.
- The correlation matrix in Table 2 indicates generally low correlations, considering SGP and CPU are only weakly positively correlated, while SGP and MPL display a slight negative correlation and CPU and PML display slightly higher correlation, which suggests potential diversification benefits.
- The covariances between assets are very small, with SGP and MPL showing a slight negative covariance, consistent with their weak negative correlation.

1.2

We utilised a quadratic utility function $U(W) = E(W) - \frac{a}{2}Var(W)$ because of its analytical simplicity as the quadratic form leads to linear relationships in portfolio optimisation and is the basis of the Markowitz portfolio theory. As we subtract the $Var(W)$, we penalise higher variance, hence, we capture risk aversion. The function is concave when a is positive, as it is increasing but with a diminishing slope, which reflects the diminishing marginal utility of wealth, making it suitable for risk-averse investors.

A limitation of the quadratic utility function is that it violates the principle of non-satiation, as it implies that utility has a maximum at a certain level of wealth and then begins to decline. For the sake of simplicity, we used an initial wealth level of \$1.

The portfolio expected return was calculated as: $E(W) = \sum_{i=1}^n w_i \mu_i$ which in vector form is $E(W) = \mathbf{w}^T \boldsymbol{\mu}$

The portfolio variance was calculated as: $Var(W) = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij}$ where σ_{ij} is the covariance between asset i and j . In vector form, the variance is $Var(W) = \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}$ where $\boldsymbol{\Sigma}$ is the covariance matrix.

We need the Lagrangian because our optimisation is constrained, as the weights must sum to 1 with no borrowing or short selling. Excel's Solver function implicitly applies the Lagrangian optimisation framework, whilst maximising the objective utility function. We used Excel's solver function to find the optimal weights by using the following steps.

1. We set the target cell as the cell with the utility function with $a = 1$.
2. We set the cells to change as the weights for the 3 companies and the risk-free asset, which is simply $1 - w_{\text{risky assets}}$.
3. We provided a constraint that the total weight must sum to 1, meaning that 100% of the investor's wealth is invested and no borrowing is possible.
4. We provided another constraint that all weights must be greater than or equal to 0, as no short selling is allowed.
5. To maximise the utility, we chose the 'max' option for the target cell.
6. We used the GRG non-linear optimisation strategy as it efficiently solves smooth, differentiable nonlinear optimisation problems subject to linear and inequality constraints, such as the no-short-selling and no-borrowing conditions in this study. It iteratively searches for the set of portfolio weights that maximise utility while satisfying all constraints.

Table 4: Optimal Weights for risky and risk-free assets to maximise utility

Company	Weight
Stockland Group	0.4788
Computershare Limited	0.3537
Medibank Private Limited	0.1676
Risk Free	0.0000

This weight allocation aligns with our expectations given the statistical characteristics of the assets. This portfolio balances return and risk efficiently for a risk aversion level of $a = 1$, it favours assets with strong performance and diversification properties.

- SGP and CPU have the highest mean weekly returns, as seen in Table 1 (0.3984% and 0.4098%), while SGP and MPL offer the lowest volatility (0.0793% and 0.0759%), making them attractive from a risk-adjusted perspective.
- A relatively large allocation in SGP reflects a combination of strong expected return and low variance, whereas CPU's higher volatility limits its weight allocation despite having the highest mean return.
- MPL receives a smaller allocation due to its low risk, but still noteworthy due to its slightly negative correlation with SGP (-0.0259), which may provide very slight diversification benefits by reducing portfolio variance.
- All three risky assets outperform the weekly risk-free rate of 0.0623% in terms of their ratio of risk and return, which explains why the risk-free asset is assigned no weight.

1.3

Using the method in section 1.2, we adjusted volatility and correlation to determine the effects on the portfolio composition.

1. For the sensitivity to volatility analysis, we first began with the baseline covariance matrix estimated from historical data, then multiplied the standard deviation of SGP by a factor of 1.05 (5% increase) and kept all else the same.
2. Using the updated correlation and unchanged correlation values, we recalculated the covariance matrix using the formula $cov_{ij} = \rho_{ij}\sigma_i\sigma_j$ and re-optimised the portfolio.
3. We repeated steps 1 and 2, instead this time multiplying the standard deviation of SGP by 1.1 to get a 10% increase.
4. We repeated steps 1-3 for the other two risky assets as well to observe the sensitivities to upward volatility change.
5. We repeated steps 1-4, but using a factor of 0.95 and then 0.9 to observe the downward change. The results of these optimisations can be seen in Table 5.

1. As for the sensitivity to correlation analysis, we started with the historical covariance matrix and computed the baseline correlation matrix based on the standard deviation and covariance values, as per the following formula:

$$\rho_{ij} = \frac{cov(R_i, R_j)}{\sigma_i\sigma_j}$$

2. Then, to analyse the sensitivities, we manually adjusted all correlations ρ_{ij} to 0.02 (naturally ρ_{ii} remains unchanged $\rho_{ii} = 1$) while keeping standard deviations constant.
3. After recomputing the updated covariance matrices using $cov_{ij} = \rho_{ij}\sigma_i\sigma_j$ for each correlation level, we re-optimised the portfolio and recorded the resulting weights.
4. We repeated steps 2 and 3 for 9 more correlation values ρ_{ij} being -0.2, -0.1, -0.2, -0.5, -0.8, 0.2, 0.1, 0.2, 0.5 and 0.8, and the results are again in Table 5.

Table 5: Sensitivity Analysis results

	SGP	CPU	MPL	Risk Free	Notes
Baseline	0.4788	0.3537	0.1676	0	
Volatility +5% SGP	0.4506	0.3634	0.1860	0	SGP becomes slightly riskier → weight falls by 2.5% CPU & MPL small increase for diversification
Volatility +10% SGP	0.4246	0.3728	0.2025	0	SGP even riskier → weight reduced by 5%, CPU & MPL weights increased
Volatility -5% SGP	0.5083	0.3424	0.1493	0	SGP less risky → weight rises CPU and MPL weights fall slightly
Volatility -10% SGP	0.5401	0.3310	0.1290	0	SGP less risky → weight increased by 6.5%, MPL weight reduced
Volatility +5% CPU	0.4921	0.3233	0.1846	0	CPU slightly riskier → weight drops 3%, SGP and MPL weights increase slightly
Volatility +10% CPU	0.5042	0.2966	0.1992	0	CPU riskier → CPU weight decreased by 5%, SGP weight slightly increased
Volatility -5% CPU	0.4632	0.3867	0.1501	0	CPU less risky → weight rises 3%, SGP weight declines, MPL slightly lower
Volatility -10% CPU	0.4459	0.4242	0.1299	0	CPU less risky → CPU weight increased by 6.5%, SGP weight decreased
Volatility +5% MPL	0.4785	0.3532	0.1683	0	MPL slightly riskier but negligible changes
Volatility +10% MPL	0.4920	0.3622	0.1458	0	MPL riskier → MPL weight reduced by 2%, SGP weight increased
Volatility -5% MPL	0.4709	0.3483	0.1809	0	MPL less risky → weight rises 1.5%, SGP and CPU weights fall slightly.
Volatility -10% MPL	0.4626	0.3430	0.1944	0	MPL less risky → MPL weight increased by 3%, SGP weight decreased
Correlation -0.02	0.4524	0.3642	0.1834	0	Slight to no negative correlation → portfolio slightly more diversified, SGP weight decreases 1.5%
Correlation -0.1	0.4414	0.3623	0.1962	0	More negative correlation → CPU and MPL weights increase for diversification and SGP weight decreases 2.5%
Correlation -0.2	0.4303	0.3603	0.2094	0	Stronger negative correlation → MPL weight increases, SGP decreases over 4.5%
Correlation -0.5	0.4076	0.3558	0.2367	0	High negative correlation → MPL weight significantly increases, SGP reduces 7%
Correlation -0.8	0.3936	0.3527	0.2537	0	Very strong negative correlation → MPL strong increase, SGP reduces by 8%
Correlation 0.02	0.4588	0.3652	0.1760	0	Slight to no positive correlation → SGP weight decreases slightly (unexpected but acceptable as there is essentially no correlation), minor increases in CPU and MPL
Correlation 0.1	0.4738	0.3676	0.1586	0	Positive correlation rises → CPU weight increases MPL weight reduced

Correlation 0.2	0.4986	0.3710	0.1304	0	Stronger positive correlation → SGP weight dominates, MPL weight reduced
Correlation 0.5	0.6307	0.3693	0.0000	0	High positive correlation → MPL removed, SGP weight substantially increased (dominates portfolio)
Correlation 0.8	0.8003	0.1997	0.0000	0	Very high positive correlation → SGP heavily dominates

It is noteworthy that no portfolio has any allocation to the risk-free asset under these conditions and an initial investment of \$1. This evinces that the expected returns from the risky assets are sufficiently attractive relative to their risk.

From the results above, we find the portfolio is moderately sensitive to changes in the individual asset's volatility.

- As expected, when an asset becomes more volatile, its optimal weight in the portfolio decreases and the portfolio shifts toward relatively less risky assets.
- Conversely, when an asset becomes less volatile, its weight increases because it now offers a more attractive risk-return trade-off. For example, when SGP's volatility increased by 5%, its weight fell by approximately 2.5%, and when it increased by 10%, its weight fell by around 5%, essentially, double the impact. Similarly, when SGP's volatility decreased by 5%, its weight rose modestly, while a 10% decrease led to a much stronger increase of approximately 5.5%, reflecting a non-linear but amplified response.
- CPU displays a similar pattern, where a 5% volatility rise reduces its weight by 2.5%, but a 10% rise pushes it further down by 6%. A 10% volatility reduction increases CPU's weight by 6.5% which is over double the increase observed with a 5% volatility reduction.
- MPL experiences minor shifts for 5% volatility changes, but the 10% adjustments are more pronounced, particularly the decline to 0.1458 when volatility rises 10% and the increase to 0.1944 when volatility falls 10%.

We observe that as volatility decreases, there are relatively larger increases in the weight allocation for the asset compared to the weight reduction when volatility is increased. This asymmetric reaction evinces that the investor places greater value on risk reduction than penalising moderate increases in risk, which aligns with their risk aversion levels.

This portfolio is highly sensitive to changes in correlations as negative correlations increase diversification by increasing allocations to assets that offset each other's risk, whereas strong positive correlations lead to concentrated positions in one dominant asset, which has a relatively high return and low volatility, being SGP in this study.

- As correlation becomes more negative, SGP's weight falls steadily while MPL's weight rises, showing that diversification benefits drive a more even split in weight allocation and heavier allocations toward risk-offsetting assets.
- Conversely, when correlations turn positive, SGP becomes increasingly dominant while MPL is reduced to zero. For higher correlations of 0.5 and 0.8, MPL is even eliminated from the portfolio, likely to avoid the relatively high volatility associated with MPL, as it also does not provide diversification benefits in these portfolios.

We ultimately find that extreme changes in correlation can have a larger impact on allocation compared to moderate changes in volatility, which emphasises the benefits of diversification. Ideally, risk-averse investors would prefer a robust combination of low volatility and stronger negative correlation to attain diversification and risk mitigation.

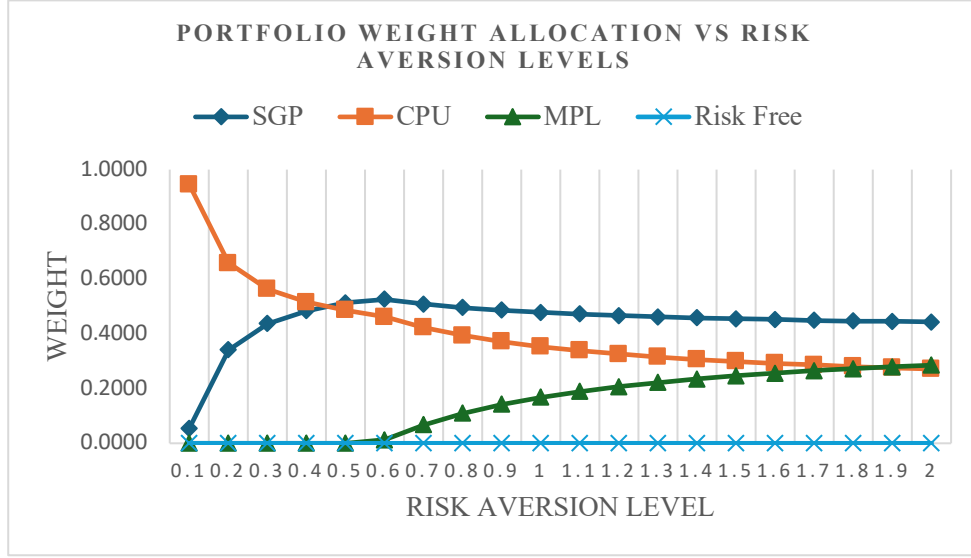
1.4

To examine how the portfolio weights change, we varied the risk aversion coefficient a from 0.1 to 2.0, increasing by increments of 0.1, while keeping all other inputs (expected returns, variances, and covariances) constant and recalculated the optimal risky asset weights for each value of a using the method in section 1.2. We plotted the weights against the corresponding risk aversion levels, as can be seen in Figure 1 below. By the quadratic utility function, we expect that as the risk aversion parameter increases (investors become more conservative), there is less weight in riskier assets. Conversely, a lower value of a represents a relatively more risk-tolerant investor, who would allocate a larger proportion of wealth to risky assets in pursuit of higher expected returns.

- We observe that with less risk aversion, the investor would invest more heavily in higher return and volatility assets, specifically, CPU in this case, with its weight exceeding 90% when $a = 0.1$.
- As a increases, CPU's weight steadily declines since the investor becomes more conservative. Conversely, the weight assigned to the more stable assets (SGP and MPL) rises as a increases.
- SGP becomes the dominant asset when a exceeds 0.5, given it has the best risk-to-return ratio, and MPL gradually gains more weight as a moves toward 2.
- CPU gradually decreases in weight due to its insufficient reward compared to the risk relative to the other assets.
- The risk-free asset consistently has no weight, indicating that the optimal risky portfolio already achieves a sufficient utility and risk-return balance.

Ultimately, we confirm that investors with low risk aversion concentrate in higher return assets, while more risk-averse investors shift toward safer, lower-volatility positions.

Figure 1: Portfolio Weight Allocation across risk aversion levels



1.5

To evaluate the risk profile of each optimal portfolio, we found the one-week 95% Value-at-Risk (VaR) for an initial portfolio value of \$100, assuming that portfolio returns are normally distributed. We used this initial wealth of \$100 to be able to better analyse the trends and observe how a higher investment may influence the weights in the risk-free asset and how that influences the value at risk and utility.

- For each level of risk aversion, we calculated new optimal portfolio weights. With the new initial wealth (W_0) of 100, we define the expected value and variance of each asset as:

$$E(r_i) = 100\mu_i \text{ and } Var(r_i) = 100^2\sigma_i^2$$
- We use this to recompute the covariance matrix as in section 1.1, then recompute the optimal weights and utility using the steps as in section 1.2, but this time for values of a between 0.1 and 2.0, increasing by increments of 0.1.

As we assume the portfolio returns follow a normal distribution, the value at risk (VaR) at a confidence level $\alpha = 0.95$ is

$$VaR_{0.95} = W_0[z_{0.95}\sqrt{var(W)} - E(W)]$$

VaR measures the maximum expected loss over the given time horizon at a 95% confidence level, providing a threshold such that actual losses are expected to exceed the amount with only a small probability. We computed the portfolio variance and VaR for each case of the risk aversion coefficient. We plotted the VaR and utility vs. the risk aversion level a as seen in Figures 2 and 3. We also plotted the weight allocation compared to the risk aversion level, similar to section 1.4, only this time with $W_0 = 100$.

A limitation here is the assumption of normally distributed returns, as financial returns often exhibit skewness, excess kurtosis, and heavy tails, which may lead to the VaR underestimating the likelihood and magnitude of extreme losses.

Figure 2: $VaR_{0.95}$ across risk aversion levels

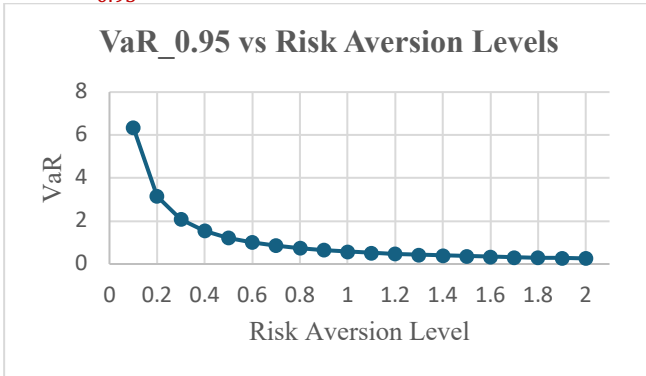
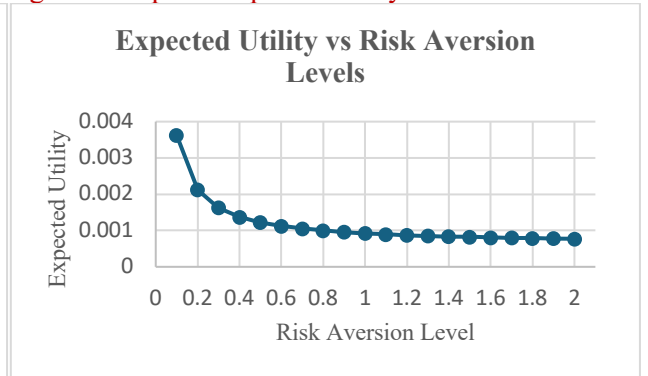


Figure 3: Expected optimal utility across risk aversion levels

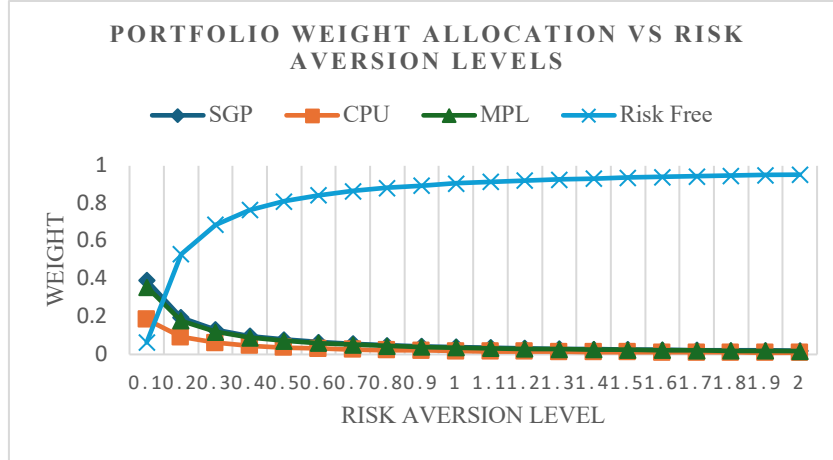


The above figures portray a clear pattern, as a increases, VaR decreases since the investor becomes more risk-averse.

- For a less risk-averse investor, the optimal portfolio contains greater allocations to risky assets, producing the largest VaR of approximately \$6. As the investor becomes moderately risk-averse, the VaR falls sharply to around \$1.2. This reflects a significant reduction in portfolio volatility as the portfolio heavily favours safer assets.

- At $a = 2$, the portfolio is dominated by the risk-free asset, resulting in a very small VaR of approximately \$0.2, which is consistent with the small exposure to market risk. This trend is also consistent with theoretical expectations, as risk aversion increases, optimal portfolios shift toward assets with lower variance, resulting in smaller downside deviations.
- The VaR curve flattens at higher levels of a , indicating diminishing marginal decreases in risk once the portfolio is already close to risk-free.

Figure 4: Portfolio weight allocation with an initial wealth of \$100 over different risk aversion levels



An interesting observation here is that a lower VaR does not necessarily correspond to a higher expected utility. Expected utility incorporates expected return and variance which is weighted by the investor's degree of risk aversion. In this study, although VaR falls as a increases, expected utility also declines. This occurs because the reduction in expected return from holding more of the risk-free asset outweighs the marginal benefit of reduced volatility. As in Figure 4, at higher risk aversion levels, the portfolio is composed of over 90% risk-free asset, hence there is a significant decrease in $E(W)$. Thus, the portfolio with the lowest VaR is not necessarily preferred by less risk-averse investors, which conveys that VaR by itself is not a sufficient measure of portfolio desirability.

Project 2

2.1

We designed a special product for moderately risk-averse investors, this product guarantees full principal protection at maturity while allowing partial participation (80%) in the upside performance of an equity index. We utilised the Binomial Option Pricing Model with 12 monthly steps to determine the fair value of the structured product at inception (time 0).

We implemented the full binomial option pricing procedure in R rather than Excel due to its better capacity to visualise the binomial tree and perform the calculations across the horizon simply and quickly.

1. We first input the parameters
 - Initial price of the index $S_0 = 100$
 - $u = 1.030835$
 - $d = 0.982593$
 - Risk-free rate $r = 5\%$
 - Participation rate $\alpha = 0.8$
 - $n = 12$
2. We calculated the time step as $\Delta t = \frac{1}{12} = \frac{T}{n}$
3. Then we found the continuously compounded monthly risk-free rate $e^{\frac{0.05}{12}}$ to find the per-step growth factor.
4. We then computed the risk-neutral probability using $q = \frac{e^{\frac{0.05}{12}} - d}{(u - d)}$.
5. For each time step, we compute the possible index levels using the following formula, which accounts for the initial index price and how many upward or downward movements the index has undergone, where i represents the number of up moves by time t . We stored the tree in a data frame for plotting and later calculations. The tree can be seen in Figure 5.

$$S_{t,i} = S_0 \times u^i d^{t-i}$$

6. We then calculated the terminal payoffs, which is calculated as

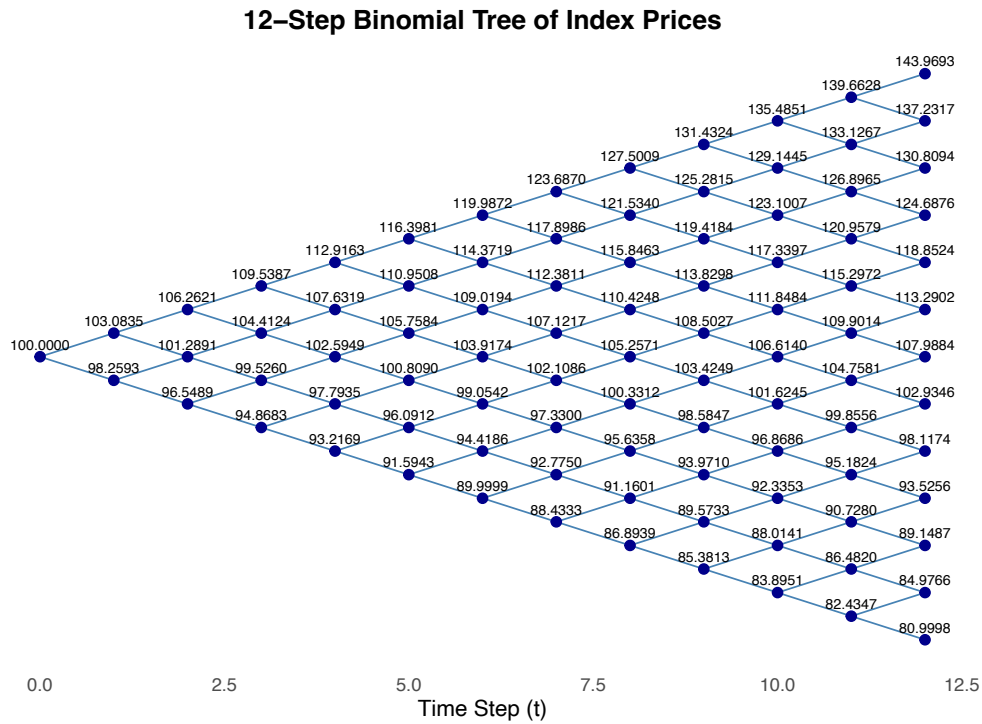
$$\text{Terminal Payoff} = \text{MAX}(100, 100(1 + 0.8(\frac{S_T}{100} - 1)))$$

This payoff structure would ensure that any node producing a value below 100 still returns the principal (principal protection) while values above 100 produce a higher payoff based on 80% of the index's return.

7. We then discounted the expected payoffs back to the present using the risk-neutral probabilities to find the initial fair price. We started from the final step and recursively computed the present value at each node with the following formula:

$$V_{t,i} = e^{-r\Delta t} [qV_{t+1,i+1} + (1 - q)V_{t+1,i}]$$

Figure 5: 12 step binomial pricing tree



The binomial tree in Figure 5 displays the possible index level over time. The final values were used to calculate the terminal payoff using:

$$\text{MAX}(100, 100(1 + 0.8(\frac{S_T}{100} - 1)))$$

Refer to Appendix 1 to see the tree for how the terminal payoff is discounted to attain the initial value.

This recursive process emulates the logic of the binomial option pricing model used for options, except here the payoff also includes a principal protection feature.

We found the fair value at time zero to be **\$100.1298**. This makes intuitive sense because the

investor receives a guaranteed return of principal plus potential upside from the index. Essentially, the product can be viewed as a combination of a zero-coupon bond that guarantees the principal and a call option that provides upside exposure.

2.2

We performed a sensitivity analysis by modifying u , d and r values, then recalculating the product price for each scenario. We varied u and d to simulate low to high volatility scenarios, during this, we kept the risk-free rate constant at 5%. We separately varied r from 1% to 9% to simulate low to high interest rate cases, keeping u and d constant at the base case.

Tables 6 and 7: Special product initial fair price as parameters are changed (base cases highlighted in the middle)

r	Price	u	d	Price
0.01	102.0296	1.010000	0.990000	99.0927
0.03	101.0047	1.020000	0.985000	99.5734
0.05	100.1298	1.030835	0.982593	100.1298
0.07	99.3856	1.050000	0.970000	101.6446
0.09	98.7511	1.060000	0.960000	102.5336

Tables 6 and 7 evince that the structured product's price is decently influenced by volatility and the risk-free rate.

- As volatility increases, the price increases, starting from 99.09, ending at 102.53. This occurs because higher volatility increases the likelihood of strong upward moves in the index, and since the losses are protected by principal protection, the upside potential dominates
- The fair price can be less than the principal of \$100 as principal protection ensures no loss at maturity, and stabilises the price under low-volatility scenarios, but it does not prevent the present value from being below \$100 when discounted at a the positive risk-free rate (5% in this case).
- Quantitatively, moving from the lowest to the highest tested volatility, the product's value increases by approximately \$3.44. This reflects the convexity in the payoff structure, which amplifies gains in high-volatility scenarios. The upside potential from larger upward moves outweighs any additional downside risk, which is mitigated by principal protection.

As for the discount rate r , this sensitivity analysis shows that the price declines as the risk-free rate increases.

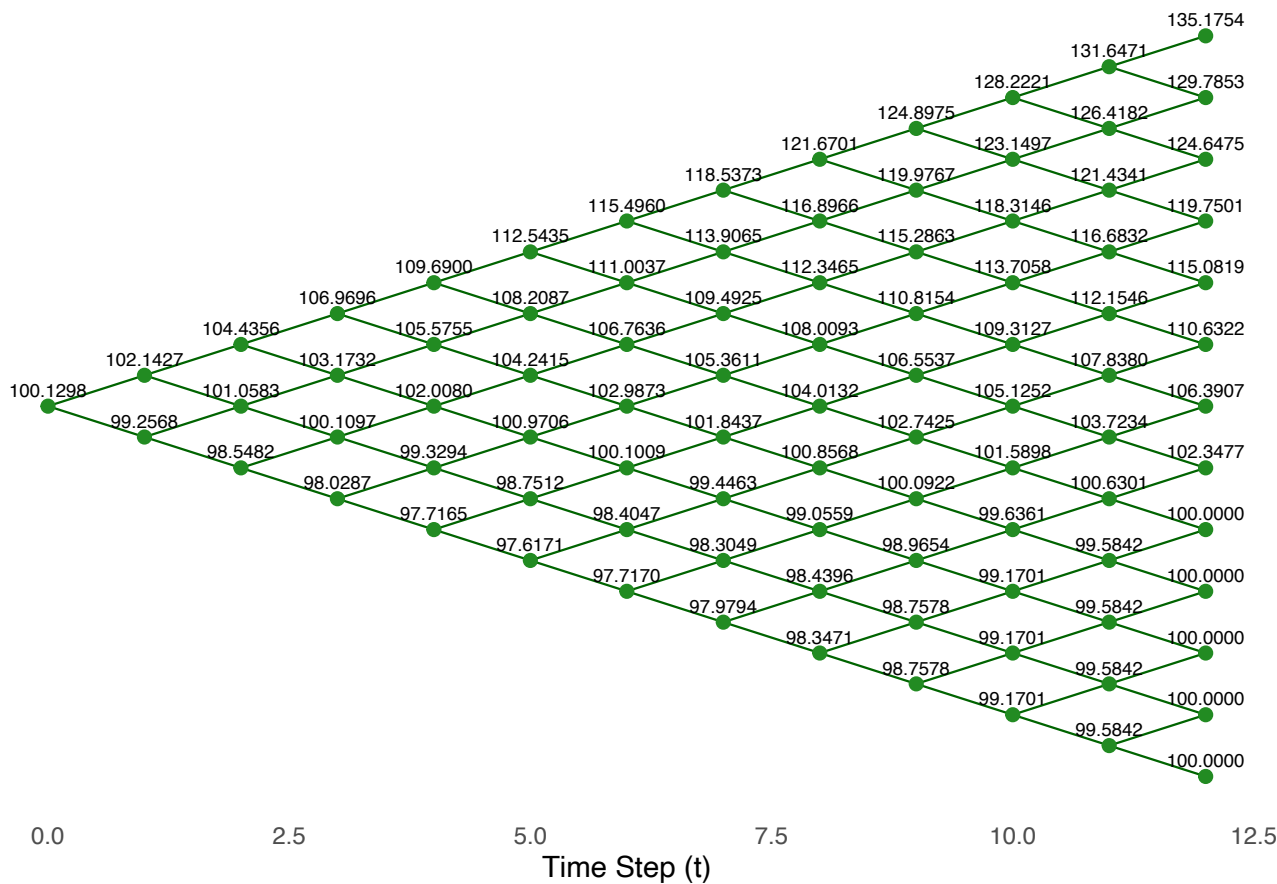
- At a low rate of 1%, the price is 102.03, while at a high rate of 9%, it falls to 98.75, a change of about \$3.28. This occurs because the present value of expected future payoffs is discounted more heavily with higher rates.
- Although higher rates slightly increase the risk-neutral probability of an up move, this effect is smaller compared to the discounting impact, which reduces the present value of future payoffs more substantially.

Qualitatively, we can view this product as a combination of a zero-coupon bond and a call option. The bond component determines the sensitivity to the interest rate, where higher rates reduce the present value, and the option component drives the sensitivity to volatility, where convexity and downside truncation amplify the product's response to increased volatility.

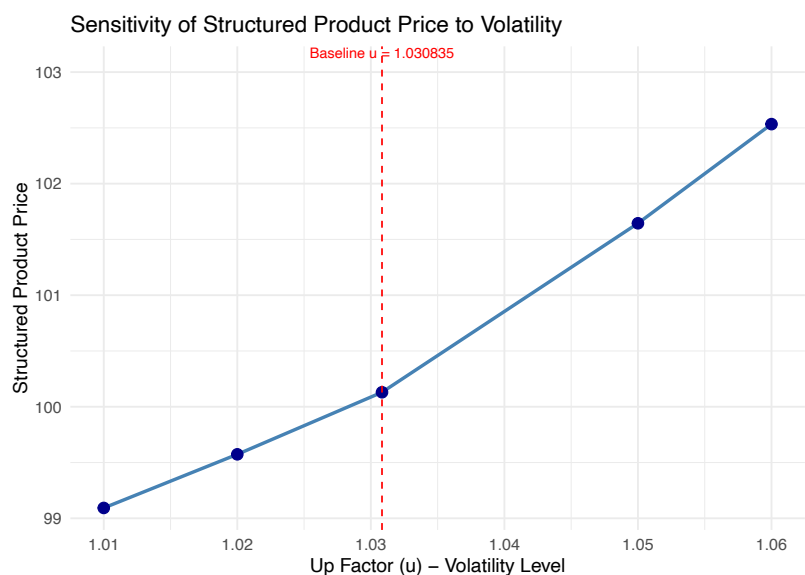
Appendix

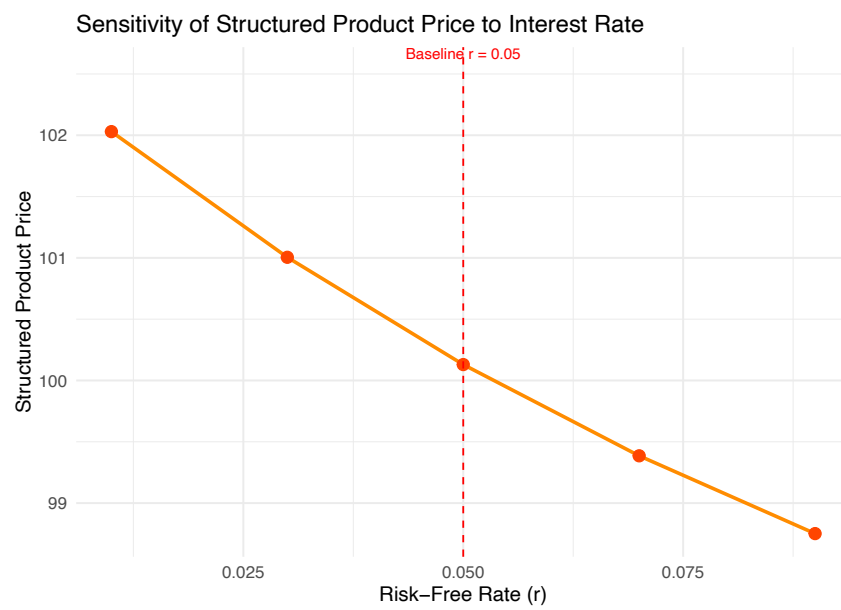
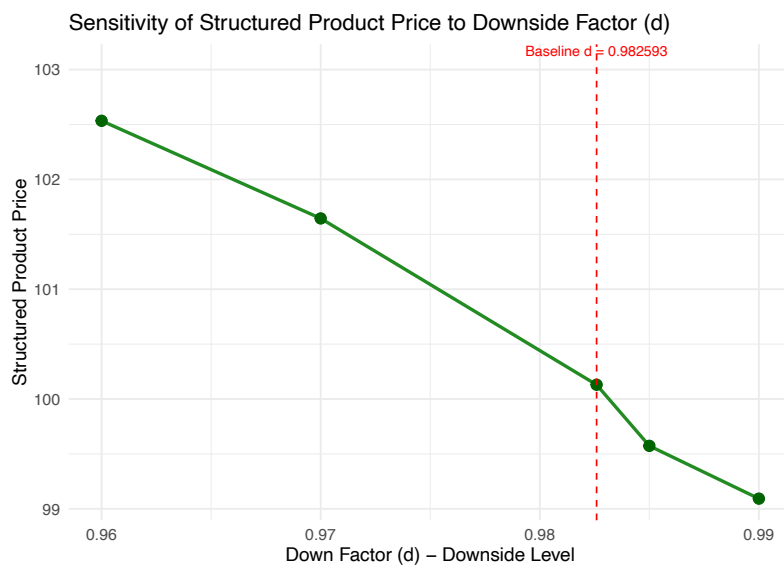
Appendix 1: Discount Binomial tree for the price of special produce

Backward-Discounted Binomial Tree of Expected Payoffs



Appendix 2: Sensitivity of product in project 2 to changes in the parameter.





AI Appendix

How do I make a binomial tree structure in R to run a loop to ultimately find the price of a special product?
How do I make a loop in R?
How do I make an ifelse statement in R?
How would I visualise a binomial tree in R?
How to set parameters in R?
How do we use Excel's solver and it's Lagrangian features?

R Code – Data and Project 2

```
# =====
# Download weekly stock data for ASX companies
# =====

library(quantmod)
library(xts)

# 1. Define tickers (ASX-listed stocks use '.AX' suffix)
tickers <- c("SGP.AX", "CPU.AX", "MPL.AX")

# 2. Download data from Yahoo Finance
getSymbols(tickers, src = "yahoo",
           from = "2022-07-04",
           to = "2025-07-04")

# 3. Function to convert daily data to weekly adjusted close
process_weekly <- function(symbol) {
  data <- get(symbol)
  # Convert to weekly adjusted closing prices
  adj_weekly <- apply.weekly(data[, paste0(symbol, ".Adjusted")], last)
  colnames(adj_weekly) <- paste0(symbol, "_WeeklyAdjusted")
  adj_weekly
}

# 4. Apply the function to each ticker
weekly_prices_list <- lapply(tickers, process_weekly)
names(weekly_prices_list) <- tickers

# 5. Merge all weekly adjusted prices
weekly_prices <- do.call(merge, weekly_prices_list)

# 6. (Optional) Inspect the first few rows
head(weekly_prices)

# 7. (Optional) Plot weekly adjusted prices
chartSeries(weekly_prices$SGP.AX_WeeklyAdjusted,
            name = "SGP Weekly Adjusted Close",
            theme = chartTheme("white"))
chartSeries(weekly_prices$CPU.AX_WeeklyAdjusted,
            name = "CPU Weekly Adjusted Close",
            theme = chartTheme("white"))
chartSeries(weekly_prices$MPL.AX_WeeklyAdjusted,
            name = "MPL Weekly Adjusted Close",
            theme = chartTheme("white"))

# 8. Save weekly adjusted prices to CSV
write.csv(data.frame(Date = index(weekly_prices),
                    coredata(weekly_prices)),
          "ASX_SGP_CPU_MPL_weekly_prices.csv", row.names = FALSE)

# -----
# Project II: Structured Product with Principal Protection
# Binomial Model Pricing + Tree Visualization + Sensitivity
# -----

library(ggplot2)
library(dplyr)

# --- PARAMETERS ---
S0 <- 100    # initial index price
W0 <- 100    # initial investment
alpha <- 0.8 # participation rate
r <- 0.05    # risk-free rate (continuous)
T <- 1       # maturity (years)
n <- 12      # monthly steps
dt <- T / n  # time step size

u <- 1 + 0.030835 # up move = +3.0835%
d <- 1 - 0.017407 # down move = -1.7407%
Tanvir Randhawa z5481573
```

```

# --- RISK-NEUTRAL PROBABILITY ---
q <- (exp(r * dt) - d) / (u - d)

# --- BUILD THE INDEX PRICE TREE ---
tree <- data.frame()
for (t in 0:n) {
  for (i in 0:t) {
    S <- S0 * u^i * d^(t - i)
    y <- i - t / 2
    tree <- rbind(tree, data.frame(t = t, i = i, S = S, y = y))
  }
}

# --- COMPUTE TERMINAL PAYOFFS (AT T = 12) ---
tree <- tree %>%
  mutate(payload = ifelse(t == n, pmax(W0, W0 * (1 + alpha * (S - S0) / S0)), NA))

# --- BACKWARD DISCOUNTING (RISK-NEUTRAL EXPECTATION) ---
for (t in (n - 1):0) {
  for (i in 0:t) {
    up <- tree$payload[tree$t == t + 1 & tree$i == i + 1]
    down <- tree$payload[tree$t == t + 1 & tree$i == i]
    tree$payload[tree$t == t & tree$i == i] <- exp(-r * dt) * (q * up + (1 - q) * down)
  }
}

tree$S <- round(tree$S, 4)
tree$payload <- round(tree$payload, 4)

price_0 <- tree$payload[tree$t == 0 & tree$i == 0]
cat("Price of structured product at time 0:", price_0, "\n\n")

price_structured <- function(u, d, r) {
  q <- (exp(r * dt) - d) / (u - d) # risk-neutral probability
  i <- 0:n # number of up moves
  ST <- S0 * u^i * d^(n - i) # terminal stock prices

  # terminal payoff with principal protection + participation
  WT <- pmax(W0, W0 * (1 + alpha * (ST - S0) / S0))

  # expected payoff under risk-neutral measure
  expected_payload <- sum(WT * dbinom(i, n, q))
  discounted_PV <- expected_payload * exp(-r * T)

  return(round(discounted_PV, 4))
}
# -----
# BACKWARD-DISCOUNTED PAYOFF TREE PLOT
# -----

discount_tree <- tree %>%
  mutate(label = sprintf("%.4f", payload))

edges_discount <- discount_tree %>%
  filter(t < n) %>%
  mutate(
    x = t,
    ystart = y,
    xend = t + 1,
    yup = y + 0.5,

```

```

    ydown = y - 0.5
  )

# Plot backward-discounted payoff tree
ggplot() +
  geom_segment(data = edges_discount, aes(x = x, y = ystart, xend = xend, yend = yup),
    linewidth = 0.4, color = "darkgreen") +
  geom_segment(data = edges_discount, aes(x = x, y = ystart, xend = xend, yend = ydown),
    linewidth = 0.4, color = "darkgreen") +
  geom_point(data = discount_tree, aes(x = t, y = y), size = 2, color = "forestgreen") +
  geom_text(data = discount_tree, aes(x = t, y = y, label = label),
    vjust = -0.7, size = 2.6, color = "black") +
  theme_minimal() +
  labs(
    title = "Backward-Discounted Binomial Tree of Expected Payoffs",
    x = "Time Step (t)",
    y = ""
  ) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank(),
    panel.grid = element_blank()
  )

```

```

# -----
# TREE PLOT (Index Tree)
# -----

```

```

edges <- tree %>%
  filter(t < n) %>%
  mutate(
    x = t,
    ystart = y,
    xend = t + 1,
    yup = y + 0.5,
    ydown = y - 0.5
  )

ggplot() +
  geom_segment(data = edges, aes(x = x, y = ystart, xend = xend, yend = yup),
    linewidth = 0.4, color = "steelblue") +
  geom_segment(data = edges, aes(x = x, y = ystart, xend = xend, yend = ydown),
    linewidth = 0.4, color = "steelblue") +
  geom_point(data = tree, aes(x = t, y = y), size = 2, color = "darkblue") +
  geom_text(data = tree, aes(x = t, y = y, label = sprintf("%.4f", S)),
    vjust = -0.7, size = 2.6) +
  theme_minimal() +
  labs(
    title = "12-Step Binomial Tree of Index Prices",
    x = "Time Step (t)",
    y = ""
  ) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank(),
    panel.grid = element_blank()
  )

```

```

# -----
# Sensitivity to Volatility (u, d)
# -----

```



```

vol_scenarios <- data.frame(
  u = c(1.01, 1.02, 1.030835, 1.05, 1.06),
  d = c(0.99, 0.985, 0.982593, 0.97, 0.96)
)

vol_scenarios$Price <- mapply(price_structured,
                             vol_scenarios$u,
                             vol_scenarios$d,
                             MoreArgs = list(r = r))

cat("Sensitivity to Volatility (u,d):\n")
print(vol_scenarios)

# -----
# Sensitivity to Risk-Free Rate
# -----
r_scenarios <- c(0.01, 0.03, 0.05, 0.07, 0.09)
r_values <- sapply(r_scenarios, function(rr) price_structured(u, d, rr))

cat("Sensitivity to Risk-Free Rate:\n")
print(data.frame(r = r_scenarios, Price = r_values))

# --- Plot: Volatility Sensitivity ---
ggplot(vol_scenarios, aes(x = u, y = Price)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_point(size = 3, color = "darkblue") +
  geom_vline(xintercept = 1.030835, linetype = "dashed", color = "red") +
  annotate("text", x = 1.030835, y = max(vol_scenarios$Price) + 0.5,
         label = "Baseline u = 1.030835", color = "red", size = 3, vjust = -1) +
  labs(
    title = "Sensitivity of Structured Product Price to Volatility",
    x = "Up Factor (u) — Volatility Level",
    y = "Structured Product Price"
  ) +
  theme_minimal(base_size = 12)

# -----
# Sensitivity to Risk-Free Rate
# -----
r_scenarios <- c(0.01, 0.03, 0.05, 0.07, 0.09)
r_values <- sapply(r_scenarios, function(rr) price_structured(u, d, rr))
rate_df <- data.frame(r = r_scenarios, Price = r_values)

cat("Sensitivity to Risk-Free Rate:\n")
print(rate_df)

# --- Plot: Interest Rate Sensitivity ---
ggplot(rate_df, aes(x = r, y = Price)) +
  geom_line(color = "darkorange", linewidth = 1) +
  geom_point(size = 3, color = "orangered") +
  geom_vline(xintercept = 0.05, linetype = "dashed", color = "red") +
  annotate("text", x = 0.05, y = max(rate_df$Price) + 0.5,
         label = "Baseline r = 0.05", color = "red", size = 3, vjust = -1) +
  labs(
    title = "Sensitivity of Structured Product Price to Interest Rate",
    x = "Risk-Free Rate (r)",
    y = "Structured Product Price"
  ) +
  theme_minimal(base_size = 12)

```

Excel Output – Project 1

Data Layout

A	B	C	D	E	F	G	H	I
Date	SGP AX Weekly Adjusted	CPU AX Weekly Adjusted	MPL AX Weekly Adjusted	SGP Return	CPU Return	MPL Return	Risk Free Rate	
15/7/2022	3.20089817	22.97409058	2.914717197				0.00039	
22/7/2022	3.217744827	23.74747849	2.957962513	0.005263	0.033663	0.014837	0.00039	
29/7/2022	3.243015289	22.72842789	2.949313641	0.007853	-0.042912	-0.002924	0.00039	
5/8/2022	3.268285751	23.15606689	2.992558718	0.007792	0.018815	0.014663	0.00039	
12/8/2022	3.217744827	21.24534798	2.992558718	-0.015464	-0.082515	0.000000	0.00039	
19/8/2022	3.141934395	22.47993469	3.156889915	-0.023560	0.058111	0.054913	0.00039	
26/8/2022	3.091394186	22.69192314	3.200135231	-0.016086	0.009430	0.013699	0.00039	
2/9/2022	2.981889248	22.34168243	3.156889915	-0.035423	-0.015435	-0.013514	0.00039	
9/9/2022	3.0155828	22.69192314	3.124586821	0.011299	0.015677	-0.010233	0.00039	
16/9/2022	2.906078577	23.97306442	3.133413315	-0.036313	0.056458	0.002825	0.00039	
23/9/2022	2.771303892	22.86704254	3.062801123	-0.046377	-0.046136	-0.022535	0.00039	
2/10/2022	2.754457235	22.61818886	3.071627855	-0.006079	-0.010883	0.002882	0.00039	
9/10/2022	2.754457235	23.55830574	3.115760088	0.000000	0.041565	0.014368	0.00044	
16/10/2022	2.720763683	23.11590004	3.001015663	-0.012232	-0.018779	-0.036827	0.00044	
23/10/2022	2.804997444	22.64583588	3.093693972	0.030960	-0.020335	0.030882	0.00044	
30/10/2022	3.032429695	23.29101563	2.480251312	0.081081	0.028490	-0.198288	0.00044	
6/11/2022	2.931349039	23.71499443	2.497904301	-0.033333	0.018204	0.007117	0.00044	
13/11/2022	3.099817038	23.73342896	2.47142458	0.057471	0.000777	-0.010601	0.00044	
20/11/2022	3.150357723	25.35559464	2.515557051	0.016304	0.068349	0.017857	0.00044	
27/11/2022	3.192474604	25.51228333	2.550863266	0.013369	0.006180	0.014035	0.00044	
4/12/2022	3.28513217	24.83945274	2.542036772	0.029024	-0.026373	-0.003460	0.00044	
11/12/2022	3.259861946	24.24957275	2.612648726	-0.007692	-0.023748	0.027778	0.00044	
18/12/2022	3.234591961	24.33252335	2.586169481	-0.007752	0.003421	-0.010135	0.00044	
22/12/2022	3.158781052	24.24957275	2.60382247	-0.023438	-0.003409	0.006826	0.00044	
29/12/2022	3.157316446	24.14818954	2.60382247	-0.000464	-0.004181	0.000000	0.00052	
8/1/2023	3.166013718	22.35089684	2.568516254	0.002755	-0.074428	-0.013559	0.00052	
15/1/2023	3.331272602	21.74258423	2.550863266	0.052198	-0.027216	-0.006873	0.00052	
22/1/2023	3.383460283	22.30481339	2.550863266	0.015666	0.025858	0.000000	0.00052	
29/1/2023	3.444344997	22.01909256	2.559689999	0.017995	-0.012810	0.003460	0.00052	
5/2/2023	3.548718691	22.6642704	2.62147522	0.030303	0.029301	0.024138	0.00052	
12/2/2023	3.392157555	22.80252457	2.736220121	-0.044118	0.006100	0.043771	0.00052	
19/2/2023	3.383460283	22.2218647	2.700913906	-0.002564	-0.025465	-0.012903	0.00052	
26/2/2023	3.270387888	23.32315254	2.912750244	-0.033419	0.049559	0.078431	0.00052	
5/3/2023	3.296481609	22.98716736	2.977916002	0.007979	-0.014406	0.022373	0.00052	
12/3/2023	3.218200922	22.14719772	2.932932377	-0.023747	-0.036541	-0.015106	0.00052	
19/3/2023	3.322574854	19.15130615	2.860958815	0.032432	-0.135272	-0.024540	0.00052	
26/3/2023	3.40085578	19.57129097	2.923935652	0.023560	0.021930	0.022012	0.00056	
30/3/2023	3.461740494	20.14993668	3.022899389	0.017903	0.029566	0.033846	0.00056	
6/4/2023	3.566114426	20.36459541	3.085876703	0.030151	0.010653	0.020833	0.00056	
14/4/2023	3.653092861	21.31655884	3.139856815	0.024390	0.046746	0.017493	0.00056	
21/4/2023	3.661791086	21.30722809	3.247817516	0.002381	-0.000438	0.034384	0.00056	
28/4/2023	3.87053895	20.90591049	3.202833891	0.057007	-0.018835	-0.013850	0.00056	
5/5/2023	3.931423426	20.299263	3.220827103	0.015730	-0.029018	0.005618	0.00056	
12/5/2023	3.896632433	20.39259338	3.166847229	-0.008849	0.004598	-0.016760	0.00056	
19/5/2023	3.861840963	20.59791946	3.166847229	-0.008929	0.010069	0.000000	0.00056	
26/5/2023	3.774862766	20.84991264	3.211830854	-0.022522	0.012234	0.014205	0.00056	
2/6/2023	3.818351746	20.68191719	3.220827103	0.011521	-0.008057	0.002801	0.00056	
9/6/2023	3.566114426	21.2605629	3.193837404	-0.066059	0.027978	-0.008380	0.00056	
16/6/2023	3.61830163	21.29789352	3.211830854	0.014634	0.001756	0.005634	0.00058	
23/6/2023	3.487834454	21.15789986	3.238820791	-0.036058	-0.006573	0.008403	0.00058	
30/6/2023	3.631860971	21.82054329	3.166847229	0.041294	0.031319	-0.022222	0.00058	
7/7/2023	3.559764624	21.67121506	3.139856815	-0.019851	-0.006843	-0.008523	0.00058	
14/7/2023	3.785065889	22.4645195	3.013902664	0.063291	0.036606	-0.040115	0.00058	
21/7/2023	3.712969303	23.32315254	3.121863365	-0.019048	0.038222	0.035821	0.00058	
28/7/2023	3.794078112	23.37915039	3.175843716	0.021845	0.002401	0.017291	0.00058	
4/8/2023	3.740005732	23.51914787	3.121863365	-0.014252	0.005988	-0.016997	0.00058	
11/8/2023	3.712969303	22.72584343	3.112866879	-0.007229	-0.033730	-0.002882	0.00058	
18/8/2023	3.712969303	23.07116318	3.121863365	0.000000	0.015195	0.002890	0.00058	
25/8/2023	3.758029938	23.40365982	3.211830854	0.012136	0.014412	0.028819	0.00058	
1/9/2023	3.803089857	23.96337318	3.265810966	0.011990	0.023916	0.016807	0.00058	
8/9/2023	3.622848988	24.4566803	3.157850266	-0.047393	0.020586	-0.033058	0.00060	

Part 1.1 and 1.2

Date	SGP.AX_WeeklyAdjusted	CPU.AX_WeeklyAdjusted	MPL.AX_WeeklyAdjusted	SGP Return	CPU Return	MPL Return	Risk Free Rate
9/3/2025	4.835632801	39.88459015	4.201778412	-0.031068	-0.034442	-0.014286	0.00080
16/3/2025	4.787179947	37.79995728	4.270339012	-0.010020	-0.052267	0.016317	0.00080
23/3/2025	4.825942516	39.16336441	4.3291049	0.008097	0.036069	0.013761	0.00080
30/3/2025	4.748417377	38.65950012	4.348693848	-0.016064	-0.012866	0.004525	0.00080
3/4/2025	4.864705086	35.51773453	4.534786701	0.024490	-0.081268	0.042793	0.00080
11/4/2025	4.874395847	36.5353508	4.3291049	0.001992	0.028651	-0.045356	0.00080
17/4/2025	5.029446125	37.73080063	4.397665024	0.031809	0.032720	0.015837	0.00081
24/4/2025	5.155424595	38.79781723	4.446637154	0.025048	0.028280	0.011136	0.00081
2/5/2025	5.378309727	38.90649414	4.61314106	0.043233	0.002801	0.037445	0.00081
9/5/2025	5.436453342	39.02505112	4.61314106	0.010811	0.003047	0.000000	0.00081
16/5/2025	5.378309727	41.21835709	4.642524242	-0.010695	0.056203	0.006369	0.00081
23/5/2025	5.320165157	39.50915909	4.720879078	-0.010811	-0.041467	0.016878	0.00081
30/5/2025	5.291093826	39.73639297	4.671907425	-0.005464	0.005751	-0.010373	0.00081
6/6/2025	5.300784111	40.58605576	4.75026226	0.001831	0.021382	0.016771	0.00081
13/6/2025	5.436453342	40.16122437	4.799233913	0.025594	-0.010467	0.010309	0.00081
20/6/2025	5.426762581	38.78793335	4.779645443	-0.001783	-0.034194	-0.004082	0.00081
27/6/2025	5.409999847	38.97565079	4.877588749	-0.003089	0.004840	0.020492	0.00081
4/7/2025	5.579999924	39.23252487	4.906971931	0.031423	0.006591	0.006024	0.00081
Question 2			Mean	0.3984%	0.4098%	0.3760%	0.0623%
			Variance	0.0793%	0.1306%	0.0759%	0.0000%
			Std Dev	0.028160104	0.03613179	0.027554648	0.000121817
		Covariance		SGP	CPU	MPL	
			SGP	0.0793%	0.0041%	-0.0020%	
			CPU	0.0041%	0.1306%	0.0141%	
			MPL	-0.0020%	0.0141%	0.0759%	
		Correlation		SGP	CPU	MPL	
			SGP	1.0000	0.0402	-0.0259	
			CPU	0.0402	1.0000	0.1418	
			MPL	-0.0259	0.1418	1.0000	
		Optimisation	Weights	SGP	0.478776832		
			CPU	0.353671431			
			MPL	0.167551816			
			Risk Free	0			
			Sum	1			
					Mean		
			SGP	0.003984			
			CPU	0.004098			
			MPL	0.003760			
			Risk Free	0.000623			
			Portfolio Expected Return	0.003986638			
			Portfolio Variance	0.000393755			
			Utility	0.00378976			

Zoomed out method for part 1.3 (layout for each section is the same as 1.2)

Weight Allocations for section 1.3

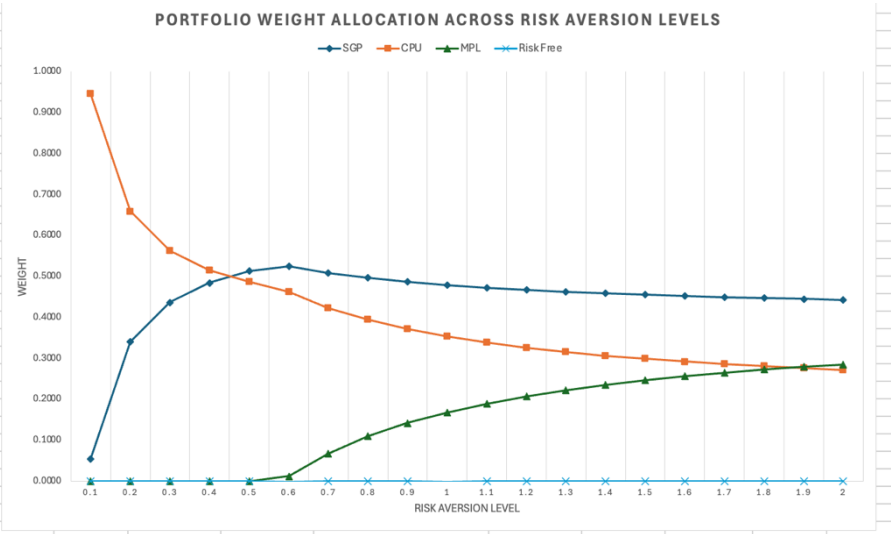
	SGP	CPU	MPL	Risk Free
	0.4246	0.3728	0.2025	0.0000
	0.5401	0.3310	0.1290	0.0000
	0.5042	0.2966	0.1992	0.0000
	0.4459	0.4242	0.1299	0.0000
	0.4920	0.3622	0.1458	0.0000
	0.4626	0.3430	0.1944	0.0000
	0.4506	0.3634	0.1860	0.0000
	0.5083	0.3424	0.1493	0.0000
	0.4921	0.3233	0.1846	0.0000
	0.4632	0.3867	0.1501	0.0000
	0.4785	0.3532	0.1683	0.0000
	0.4709	0.3483	0.1809	0.0000
	0.4588	0.3652	0.1760	0.0000
	0.4738	0.3676	0.1586	0.0000
	0.4986	0.3710	0.1304	0.0000
	0.6307	0.3693	0.0000	0.0000
	0.8003	0.1997	0.0000	0.0000
	0.4524	0.3642	0.1834	0.0000
	0.4414	0.3623	0.1962	0.0000
	0.4303	0.3603	0.2094	0.0000
	0.4076	0.3558	0.2367	0.0000
	0.3936	0.3527	0.2537	0.0000

Zoomed out method for part 1.4 (layout for each section is the same as 1.2)



Weight Allocations for section 1.4 and graph

a	SGP	CPU	MPL	Risk Free
0.1	0.0546	0.9454	0	0
0.2	0.3412	0.6588	0	0
0.3	0.4367	0.5633	0	0
0.4	0.4845	0.5155	0	0
0.5	0.5132	0.4868	0	0
0.6	0.5256	0.4622	0.0122	0
0.7	0.5088	0.4233	0.0680	0
0.8	0.4962	0.3941	0.1098	0
0.9	0.4863	0.3714	0.1423	0
1	0.4785	0.3532	0.1683	0
1.1	0.4721	0.3384	0.1896	0
1.2	0.4667	0.3260	0.2073	0
1.3	0.4622	0.3155	0.2223	0
1.4	0.4583	0.3065	0.2352	0
1.5	0.4549	0.2987	0.2463	0
1.6	0.4520	0.2919	0.2561	0
1.7	0.4494	0.2859	0.2647	0
1.8	0.4471	0.2806	0.2723	0
1.9	0.4450	0.2758	0.2792	0
2	0.4432	0.2715	0.2853	0



Zoomed out method for part 1.5 (layout for each section is the same as 1.2)



Weights, VaR and Utility for section 1.5

a	Weights				Mean	Standard Deviation	VaR_0.95 Across Risk Aversion Levels	Expected Utility Across Risk Aversion Levels
	SGP	CPU	MPL	Risk Free				
0.1	0.392500605	0.185467015	0.3580395	0.06	0.00371	0.04093	6.361368174	0.003626128
0.2	0.196250297	0.092733543	0.1790197	0.53	0.00217	0.02046	3.149510232	0.0021248
0.3	0.130833716	0.061822237	0.1193462	0.69	0.00165	0.01364	2.078889482	0.001624357
0.4	0.09812529	0.046366637	0.0895095	0.77	0.00140	0.01023	1.543579007	0.001374135
0.5	0.078500093	0.037093203	0.0716074	0.81	0.00124	0.00819	1.222390926	0.001224001
0.6	0.065416849	0.030911112	0.0596732	0.84	0.00114	0.00682	1.008272097	0.001123915
0.7	0.056071544	0.026495284	0.0511484	0.87	0.00106	0.00585	0.855326121	0.001052423
0.8	0.049062587	0.023183251	0.0447548	0.88	0.00101	0.00512	0.740615362	0.000998803
0.9	0.043610923	0.020607319	0.0397822	0.90	0.00097	0.00455	0.651396575	0.000957099
1	0.039249902	0.018546648	0.035804	0.91	0.00093	0.00409	0.580023304	0.000923737
1.1	0.035681881	0.016860653	0.032549	0.91	0.00090	0.00372	0.521626869	0.000896441
1.2	0.032708392	0.015455595	0.0298366	0.92	0.00088	0.00341	0.472962442	0.000873693
1.3	0.030192352	0.014266704	0.0275415	0.93	0.00086	0.00315	0.431784838	0.000854446
1.4	0.028035739	0.013247655	0.0255743	0.93	0.00084	0.00292	0.396489724	0.000837947
1.5	0.02616667	0.012364477	0.0238693	0.94	0.00083	0.00273	0.36590059	0.000823649
1.6	0.024531227	0.011591693	0.0223775	0.94	0.00082	0.00256	0.339135009	0.000811138
1.7	0.023088185	0.010909822	0.0210612	0.94	0.00081	0.00241	0.315518273	0.000800099
1.8	0.021805478	0.010303713	0.0198912	0.95	0.00079	0.00227	0.294525552	0.000790286
1.9	0.020657745	0.009761276	0.0188441	0.95	0.00079	0.00215	0.275740313	0.000781505
2	0.019624866	0.009273195	0.0179019	0.95	0.00078	0.00205	0.258836005	0.000773604

